

Sunil Jaat

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Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Bachelor of Technology in Artificial Intelligence

2024 – Present
CGPA: 8.30

Related Coursework

Data Structures & Algorithms , Object-Oriented Programming , Database Management Systems , Operating Systems , Computer Networks

Technical Skills

Languages	C++, Python, JavaScript, SQL, Rust
Backend Technologies	Node.js, Express.js, REST APIs, Socket.io
Databases	MongoDB, Redis
Machine Learning	PyTorch, Scikit-learn, OpenCV, YOLOv8, ByteTrack
Developer Tools	Git, Docker, Postman, Cloudinary

Experience

AI Intern Jul 2026 – Present
CogntonAI

- Working on the development of an agentic, memory-safe Next-Generation Firewall (NGFW).
- Contributing to AI-driven security workflows and backend development as part of the engineering team.

Projects

DevBoard — Full Stack Project Management Platform Live | GitHub

- Built a production-grade MERN project management platform with collaborative Kanban boards, analytics dashboards, and role-based workspaces.
- Developed scalable REST APIs featuring JWT authentication, Redis caching, BullMQ queues, Socket.io, MongoDB aggregation pipelines, and Swagger documentation.
- Improved backend scalability through asynchronous job processing, database indexing, caching, and optimized API design.

KV-Server — Multithreaded In-Memory Key-Value Store (Rust) GitHub

- Built a multithreaded in-memory key-value store in Rust using raw TCP sockets, enabling concurrent client connections through thread-safe shared state with `Arc<Mutex<HashMap<...>>>`.
- Implemented durable JSON-based persistence with automatic recovery on restart and key expiration (TTL) using lazy expiration and a background reaper thread.
- Designed a custom TCP protocol supporting `SET`, `GET`, `DEL`, `LIST`, and `EXPIRE` operations with concurrent request handling.

Tactics Cam — Football Tactical Intelligence Pipeline GitHub

- Built a computer vision pipeline using YOLOv8, ByteTrack, OpenCV, and K-Means clustering for player detection, tracking, and tactical analysis.
- Fine-tuned YOLOv8 achieving **0.815 mAP@50** and implemented persistent multi-object tracking on broadcast football footage.
- Developed team assignment through jersey-color clustering and generated tactical analytics including team-shape estimation and zone occupancy.

Achievements

- LeetCode **Knight** with a peak contest rating of **2033**, ranked in the **Top 2.17%** globally.
- Solved **1000+** algorithmic problems on LeetCode, Codeforces and CodeChef.
- Codeforces **Pupil** (Max. **1308**).
- Published co-author with two research papers accepted at the **ACL BHASHA 2025 Workshop**.
- Executive Member of the **ACM Student Chapter**, SVNIT Surat.